

Isolating Chemicals

The discovery of the medicinal value of foxglove (*Digitalis purpurea*, p. 202) is another case where traditional herbal knowledge led to a major advance in medicine. Dr. William Withering (1741–1799), a conventionally trained doctor with a long interest in medicinal plants, started to investigate foxglove after encountering a family recipe for curing dropsy (water retention). He found that in some regions of England, foxglove was traditionally used to treat this condition, which is often one of the indications of a failing heart. In 1785, he published *Account of the Foxglove*, documenting dozens of carefully recorded case histories, and showing how foxglove's powerful (and potentially dangerous) active constituents, now known as cardiac glycosides, made it a valuable plant medicine for dropsy. Cardiac glycosides remain in common use to the present day. Yet despite this clearcut example of the possibilities inherent in a marriage of herbal medicine and scientific method, conventional medicine was to take another path in the 19th century.

Laboratory versus Nature

From the early 19th century, the chemical laboratory began to regularly supplant Mother Nature as the source of medicines. In 1803, narcotic alkaloids were isolated from the opium poppy (*Papaver somniferum*, p. 244). A year later, inulin was extracted from elecampane (*Inula helenium*, p. 107). In 1838, salicylic acid, a chemical forerunner of aspirin, was isolated from white willow bark (*Salix alba*, p. 129), and was first synthesized in the laboratory in 1860. From this point on, herbal



Opium poppy, native to Asia, yields a resin that has long been smoked for its narcotic effect. The main active constituent, morphine, was first isolated in the laboratory in 1803 and is used to relieve pain.

medicine and biomedicine were to take separate paths. Aspirin, an entirely new chemical formulation, was first developed in Germany in 1899. But this was still an early step. For the time being, the influence of the universities, medical schools, and laboratories of Europe would remain limited, and herbal medicine would prevail as the predominant form of treatment for most people around the world.

New Frontiers, New Herbal Medicines

Wherever Europeans settled during the great migrations of the 18th and 19th centuries—North America, South America, southern Africa, or Australia—much of the European medicine familiar from home was either unavailable or prohibitively expensive. Settlers came to learn that native peoples were a wellspring of information about the medicinal virtues of indigenous plants. For example, European settlers in southern Africa learned about the diuretic properties of buchu (*Barosma betulina*, p. 71) from native peoples; and Australian settlers came to understand the remarkable antiseptic properties of tea tree (*Melaleuca alternifolia*, p. 112) from observing the medicinal practices of the Aborigines. Mexican herbal medicine as it exists today is a blend of Aztec, Mayan, and Spanish herbs and practices.

In North America, native herbalists were particularly adept at healing external wounds and bites—being superior in many respects to their European counterparts in this area of medicine. This is not surprising, given the range of highly

In the 18th century, the physician William Withering documented foxglove's ability to restore a failing heart.